AI ASSISTED CODING

Lab Assignment-17.1

Name : T.Shylasri

H.T.NO: 2303A51876

Batch-14(LAB-17)

Date: 23-03-2026

Lab 17 – AI for Data Processing: Data Cleaning and Preprocessing Scripts

Lab Objectives:

• Learn how to clean raw datasets using AI-assisted Python scripting.

• Apply preprocessing techniques such as handling missin values, encoding categorical data, and normalization.

• Automate repetitive data-cleaning tasks with AI-generated code.

• Understand how preprocessing impacts model performance.

Task 1 – Customer Data Cleaning

Task:

Use AI to generate a Python script for cleaning a customer dataset.

Instructions:

• Handle missing values in columns (age, income, city).

• Remove duplicate records.

• Standardize text values (e.g., city names in lowercase).

• Encode categorical variables (gender, city).

Sample Code:

```
import pandas as pd

data = pd.read_csv("shopping_trends.csv")

print(data.head())
```

Dataset link:

https://www.kaggle.com/datasets/iamsouravbanerjee/customer-shopping-trends-dataset?select=shopping_trends.csv

Expected Output:

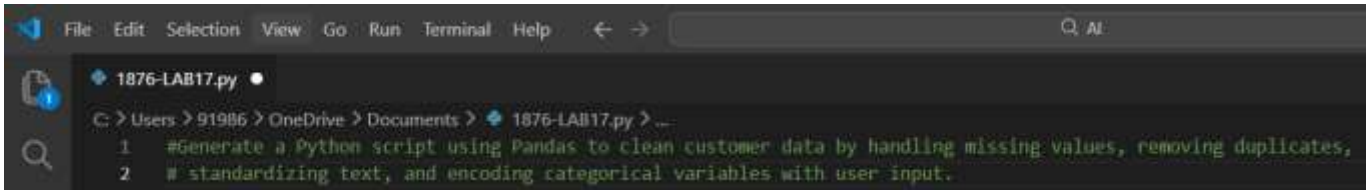
• A cleaned Pandas data frame ready for analysis.

Customer Data Cleaning

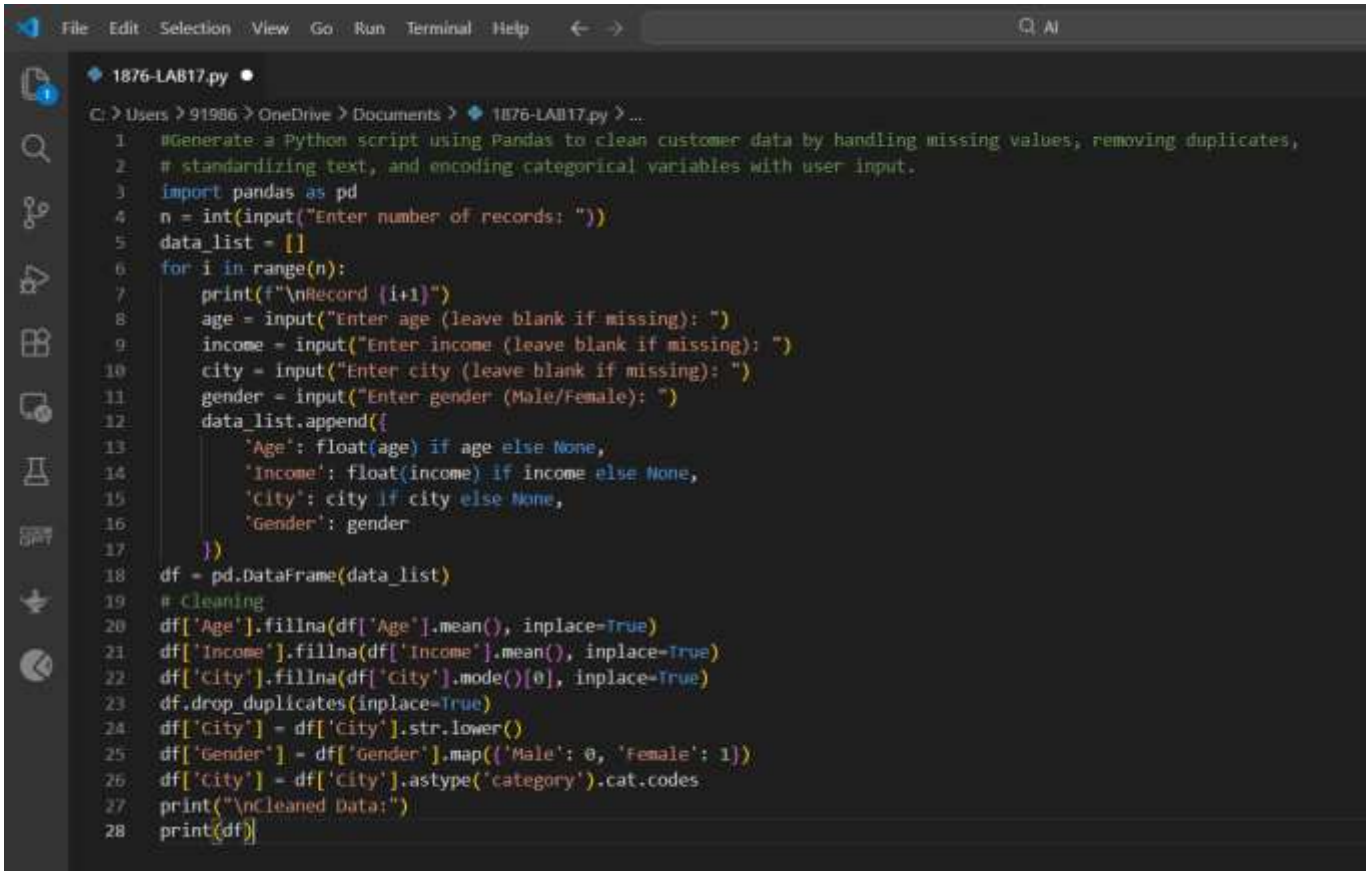
AI Prompt

#Generate a Python script using Pandas to clean customer data by handling missing values, removing duplicates, standardizing text, and encoding categorical variables with user input.

Prompt Screenshot



Python Code



Output

```
PS C:\Users\91980\OneDrive\Desktop\AI> & "C:\Program Files\Python313\python.exe" c:/Users/91980/OneDrive/Documents/1876-LA017.py
Enter number of records: 4

Record 1
Enter age (leave blank if missing): 60
Enter income (leave blank if missing): 40000
Enter city (leave blank if missing): Hanumkonda
Enter gender (Male/Female): Female

Record 2
Enter age (leave blank if missing): 47
Enter income (leave blank if missing): 150000
Enter city (leave blank if missing): Hasanparthy
Enter gender (Male/Female): Male

Record 3
Enter age (leave blank if missing): 21
Enter income (leave blank if missing): 100000
Enter city (leave blank if missing): America
Enter gender (Male/Female): Female

Record 4
Enter age (leave blank if missing): 24
Enter income (leave blank if missing): 150000
Enter city (leave blank if missing): London
Enter gender (Male/Female): Female
c:\Users\91980\OneDrive\Documents\1876-LA017.py:20: ChainedAssignmentError: A value is being set on a copy of a DataFrame or Series through chained assignment using an inplace method.
Such inplace method never works to update the original DataFrame or Series, because the intermediate object on which we are setting values always behaves as a copy (due to Copy-on-Write).

  df['Age'].fillna(df['Age'].mean(), inplace=True)
c:\Users\91980\OneDrive\Documents\1876-LA017.py:21: ChainedAssignmentError: A value is being set on a copy of a DataFrame or Series through chained assignment using an inplace method.
Such inplace method never works to update the original DataFrame or Series, because the intermediate object on which we are setting values always behaves as a copy (due to Copy-on-Write).

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' instead, to perform the operation inplace on the original object, or try to avoid an inplace operation using 'df[col
] = df[col].method(value)'.

See the documentation for a more detailed explanation: https://pandas.pydata.org/pandas-docs/stable/user_guide/copy_on_write.html
  df['Income'].fillna(df['Income'].mean(), inplace=True)
c:\Users\91980\OneDrive\Documents\1876-LA017.py:22: ChainedAssignmentError: A value is being set on a copy of a DataFrame or Series through chained assignment using an inplace method.
Such inplace method never works to update the original DataFrame or Series, because the intermediate object on which we are setting values always behaves as a copy (due to Copy-on-Write).

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' instead, to perform the operation inplace on the original object, or try to avoid an inplace operation using 'df[col
] = df[col].method(value)'.

See the documentation for a more detailed explanation: https://pandas.pydata.org/pandas-docs/stable/user_guide/copy_on_write.html
  df['City'].fillna(df['City'].mode()[0], inplace=True)

Cleaned Data:
   Age   Income  City  Gender
0  60.0  40000.0     1       1
1  47.0 150000.0     2       0
2  21.0 100000.0     0       1
3  24.0 150000.0     3       1
```

Observation

- Missing values were filled using mean and mode
- Duplicate records were removed
- City names converted to lowercase
- Categorical data encoded into numbers

Explanation

- Pandas functions like fillna(), drop_duplicates() were used
- Mean applied for numeric data, mode for categorical
- Encoding converts text data into numeric format

Justification

- Cleaning improves data quality and consistency
- Removes errors caused by missing or duplicate values
- Encoding is required for machine learning models

Conclusion

- AI simplifies data preprocessing tasks
- Cleaned dataset is ready for analysis
- Proper preprocessing improves accuracy of results

Task 2 – Sales Data Preprocessing

Task:

Preprocess sales transaction data using AI assistance.

Instructions:

• Convert date columns into datetime format.

• Create new features (month, year, day of week).

• Normalize sales amount column.

• Detect and handle outliers in dataset.

Sample Code:

```
import pandas as pd

data = pd.read_csv("sales_data.csv")

print(data.head())
```

Dataset link:

https://www.kaggle.com/datasets/vinothkannaece/sales-dataset?select=sales_data.csv

Expected Output:

• A transformed dataset with time-based features and normalized values.

Sales Data Preprocessing

AI Prompt

Generate a Python Pandas script to preprocess sales data by converting dates, creating time-based features, normalizing sales, and handling outliers with user input.

Prompt Screenshot

Python Code

Output

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS    POSTMAN CONSOLE

PS C:\Users\91986\OneDrive\Desktop\AI> & "C:/Program Files/Python313/python.exe" c:/Users/91986/OneDrive/Documents/1876-LAB17.py
Enter number of sales records: 4

Record 1
Enter date (YYYY-MM-DD): 2005-12-31
Enter sales amount: 100000

Record 2
Enter date (YYYY-MM-DD): 1978-07-28
Enter sales amount: 150000

Record 3
Enter date (YYYY-MM-DD): 1985-08-24
Enter sales amount: 20000

Record 4
Enter date (YYYY-MM-DD): 2011-04-24
Enter sales amount: 20000

Processed Data:
        Date     Sales  Month  Year DayOfWeek
0 2005-12-31  0.615385     12  2005  Saturday
1 1978-07-28  1.000000      7  1978    Friday
2 1985-08-24  0.000000      8  1985  Saturday
3 2011-04-24  0.000000      4  2011    Sunday
```

Observation

- Date converted into datetime format
- New features (month, year, day) created
- Sales normalized between 0 and 1
- Outliers detected and removed

Explanation

- pd.to_datetime() used for date conversion
- .dt functions extract time features
- Min-Max scaling normalizes values
- IQR method identifies outliers

Justification

- Time features help in trend analysis
- Normalization improves model performance
- Removing outliers prevents skewed results

Conclusion

- AI simplifies preprocessing tasks
- Data becomes structured and analysis-ready
- Preprocessing improves accuracy and insights

Task 3 – Healthcare Data Preparation

Task:

Clean and preprocess a healthcare dataset using AI scripts.

Instructions:

• Handle missing values in blood_pressure, cholesterol.

• Encode categorical features

• Scale numerical features (min-max or standard scaling).

• Split dataset into training and testing sets.

Dataset link :

https://www.kaggle.com/datasets/abdallaahmed77/healthcare-risk-factors-dataset

Expected Output:

• A preprocessed dataset ready for machine learning models.

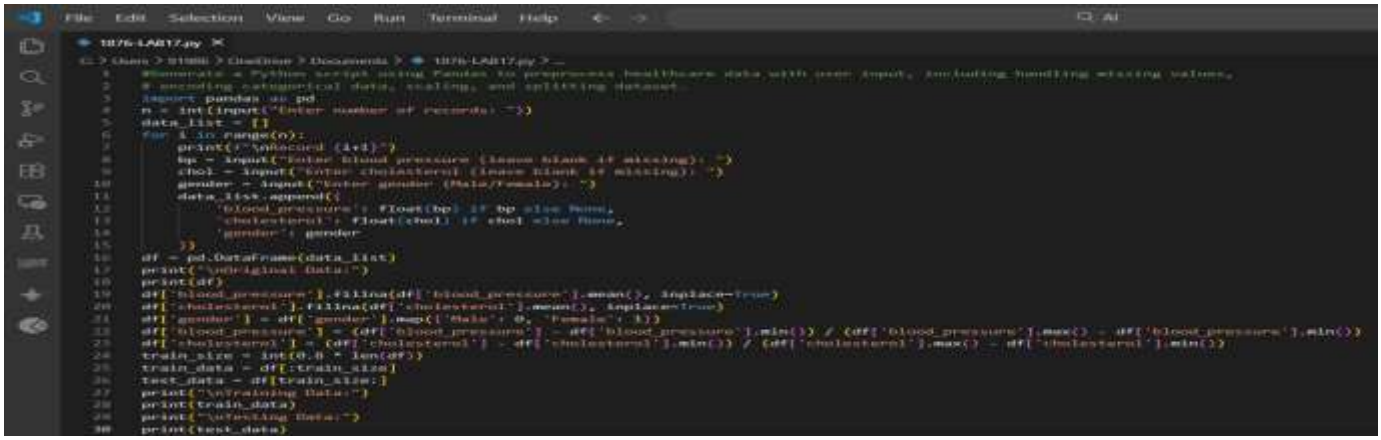
Healthcare Data Preparation

AI Prompt

#Generate a Python script using Pandas to preprocess healthcare data with user input, including handling missing values, encoding categorical data, scaling, and splitting dataset.

Prompt Screenshot

Python Code



Output

```
PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS   PYTHON DEBUG CONSOLE                                                    Python  +  ⊞  🗑  ...  ×

PS C:\Users\91986\OneDrive\Desktop\AI> & C:/Users/91986/anaconda3/python.exe c:/Users/91986/OneDrive/Documents/1876-LAB17.py
Enter number of records: 4

Record 1
Enter blood pressure (leave blank if missing): 120
Enter cholesterol (leave blank if missing): 100
Enter gender (Male/Female): Male

Record 2
Enter blood pressure (leave blank if missing): 90
Enter cholesterol (leave blank if missing): 100
Enter gender (Male/Female): Female

Record 3
Enter blood pressure (leave blank if missing): 130
Enter cholesterol (leave blank if missing): 200
Enter gender (Male/Female): Female

Record 4
Enter blood pressure (leave blank if missing): 140
Enter cholesterol (leave blank if missing): 150
Enter gender (Male/Female): Female

Original Data:
   blood_pressure  cholesterol  gender
0           120.0        100.0    Male
1            90.0        100.0  Female
2           130.0        200.0  Female
3           140.0        150.0  Female
For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col] = df[col].method(value) instead, to perform the operation inplace on the original object.


  df['blood_pressure'].fillna(df['blood_pressure'].mean(), inplace=True)
c:\Users\91986\OneDrive\Documents\1876-LAB17.py:30: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through chained assignment using an inplace method.
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are setting values always behaves as a copy.

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col] = df[col].method(value) instead, to perform the operation inplace on the original object.


  df['cholesterol'].fillna(df['cholesterol'].mean(), inplace=True)

Training Data:
   blood_pressure  cholesterol  gender
0             0.0          0.0       0
1             0.0          0.0       1
2             0.0          1.0       1

Testing Data:
   blood_pressure  cholesterol  gender
3             1.0          0.5       1
```

Observation

- Missing values handled using mean
- Categorical data encoded manually
- Features scaled without external libraries
- Dataset split manually into train and test

Explanation

- Pandas used for data processing
- Min-Max formula applied manually for scaling
- Simple slicing used for train-test split

Justification

- Avoids dependency on external libraries like sklearn
- Maintains full control over preprocessing steps
- Suitable for basic data preparation tasks

Conclusion

- Data successfully preprocessed without sklearn
- Dataset is ready for machine learning
- AI simplifies even manual preprocessing methods

Task 4 – Employee Salary Data Preprocessing

Task:

Clean and preprocess an employee salary dataset using AI scripts.

Instructions:

• Handle missing values

• Detect and remove salary outliers.

• Standardize department names (lowercase).

• Apply label encoding to job location, department.

Dataset link:

https://www.kaggle.com/code/vishantmathur/employee-salary

Expected Output:

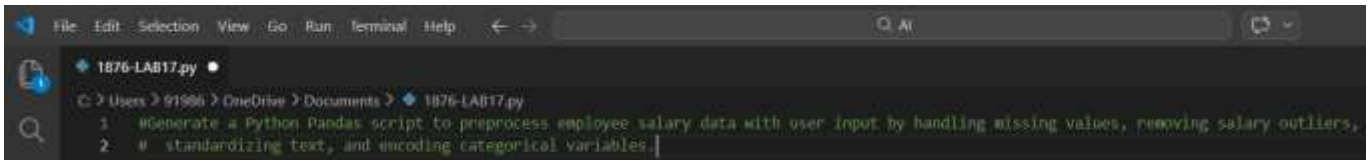
• A structured dataset ready for HR analytics.

Employee Salary Data Preprocessing

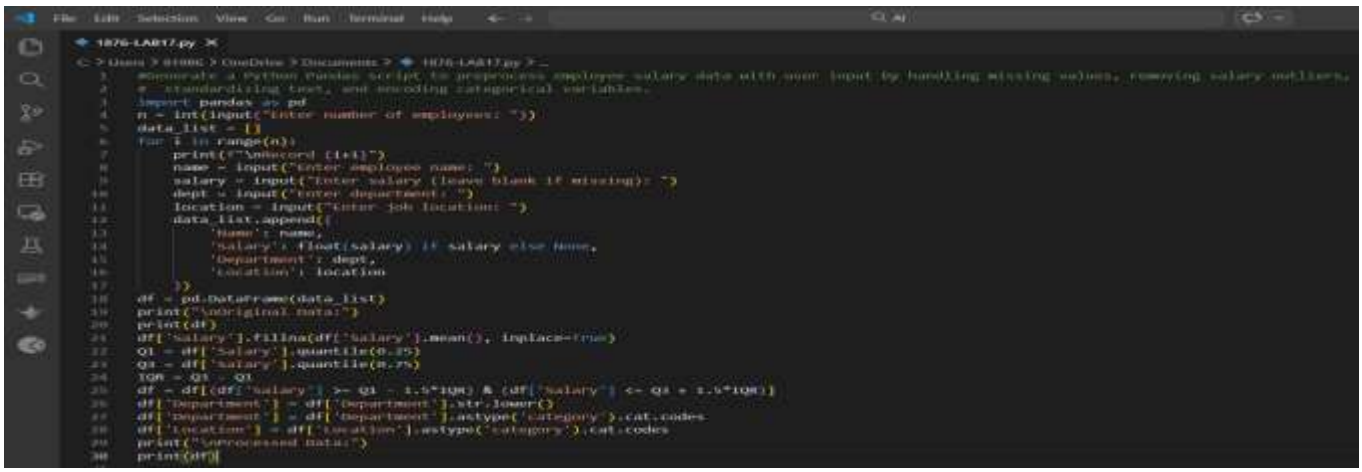
AI Prompt

#Generate a Python Pandas script to preprocess employee salary data with user input by handling missing values, removing salary outliers, standardizing text, and encoding categorical variables.

Prompt Screenshot



Python Code



Output

```
PS C:\Users\91986\OneDrive\Desktop\AI> & C:/Users/91986/anaconda3/python.exe c:/Users/91986/OneDrive/Documents/1876-LAB17.py
Enter number of employees: 4

Record 1
Enter employee name: Vasudev
Enter salary (leave blank if missing): 150000
Enter department: Maths
Enter job location: Bhupalpally

Record 2
Enter employee name: Shylasri
Enter salary (leave blank if missing): 60000
Enter department: Web Development
Enter job location: london

Record 3
Enter employee name: Harshitha
Enter salary (leave blank if missing): 10000
Enter department: Slavery
Enter job location: Beijer

Record 4
Enter employee name: Bhanuprakaja
Enter salary (leave blank if missing): 50000
Enter department: Criminal
Enter job location: Metpally

Original Data:
          Name   Salary      Department    location
0      Vasudev  150000.0           Maths  Bhupalpally
1     Shylasri   60000.0 Web Development      london
2    Harshitha   10000.0         Slavery       Beijer
c:\Users\91986\OneDrive\Documents\1876-LAB17.py:21: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through chained assignment using an inplace method.
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are setting values always behaves as a copy.

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col] = df[col].method(value) instead, to perform the operation inplace on the original object.


  df['Salary'].fillna(df['Salary'].mean(), inplace=True)

Processed Data:
          Name  Salary  Department  location
1     Shylasri  60000.0           2         1
2    Harshitha  10000.0           1         0
5  Bhanuprakaja  50000.0           0         2
```

Observation

- Missing salary values replaced using mean
- Outliers detected and removed
- Department names standardized to lowercase
- Categorical data encoded into numbers

Explanation

- Pandas used for cleaning and transformation
- IQR method used for outlier detection
- Encoding converts text to numeric form

Justification

- Cleaning improves data consistency
- Removing outliers prevents misleading analysis
- Encoding required for analytics and ML models

Conclusion

- AI simplifies preprocessing of employee data
- Dataset becomes structured and analysis-ready
- Useful for HR analytics and decision-making